The nutrient delivery investigation

Teacher guide and worked answers

Describe how absorbed nutrients and water reach body tissues and distinguish this from the waste route.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Spring | B35:C35

Describe how nutrients and water are transported in the body.

Directly develops this stated outcome with secondary-age tasks at an upper-KS2 conceptual level. Support and stretch are flexible; this exemplar is not a claim of qualification approval.

Prior learning

Know the jobs of digestion and circulation.

Trace blood through heart, lungs and body.

Success evidence

Separate digestion, absorption and transport.

Trace a soluble nutrient or water from intestine into blood and towards tissues.

Distinguish unabsorbed gut material from urine made by kidneys.

Equipment

Shared route/evidence cards, selected pupil route, pencils, optional paper tokens.

Preparation

Explain that a glucose token represents a molecule; it is not a visible crumb.

All transport diagrams are selective models, not exact anatomy or a literal sieve.

Practical care

No food, liquids, bodily samples or pupil measurements are used.

Do not prescribe how much water a pupil should drink or ask for toilet/medical disclosures.

Ready to teach alternative

Every pupil can use the supplied paper tokens and route cards. The model needs no practical materials.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Two tokens need a route
2	Retrieval	2-5	Recover the two system jobs

Slide	Stage	Minutes	Focus
3	I do	5-13	I separate breaking down from crossing
4	I do	Within stage	I follow the transport job
5	I do	Within stage	I separate two waste routes
6	We do	13-21	Delivery desk: assign the tokens
7	We do	Within stage	The incorrect shortcut
8	We do	Within stage	Choose the right exit route
9	Hinge	21-24	A water molecule is in the small intestine
10	You do	24-34	Create the delivery explanation
11	You do	Within stage	Use three different verbs
12	You do	Within stage	Check the substance at each arrow
13	Review	34-38	Trace and challenge
14	Review	Within stage	Choose a closer look
15	Exit	38-40	Three steps, three meanings

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Two tokens need a route

These are molecule models, not food crumbs. Invite a predicted route; do not mark guesses publicly.

2 Retrieval Recover the two system jobs

R1 breakdown into smaller absorbable substances; R2 blood; R3 lungs. Explain that today focuses on crossing the gut wall, not repeating the full heart route.

3 I do I separate breaking down from crossing

Think aloud: A crumb is not the transported token. I choose glucose, a small dissolved molecule. It crosses the intestinal lining into blood. The arrow shows net entry, not an empty hole. Actual absorption involves cells and transport processes; mechanism names are not required.

4 I do I follow the transport job

Shared I do time. Think aloud: Water is already a small substance; I do not need to cut it into smaller food units. Water is absorbed in small and large intestine. Most glucose from gut travels first through liver before wider circulation; detailed organ route omitted in this job map.

5 I do I separate two waste routes

Shared I do time. Trace each row separately. The urine row is simplified: ureters carry urine from kidneys to bladder and urethra carries it out. No pupil experiences are requested.

6 We do Delivery desk: assign the tokens

Tokens: large starch molecule needing digestion → break down; glucose at gut wall → absorption; dissolved glucose in blood → transport; water in the gut → absorption. No physical substances are used.

7 We do The incorrect shortcut

Shared We do time. Correct core chain: intestine absorption → blood → kidneys make urine → bladder. Do not imply all ingested water immediately becomes urine or use a fixed percentage.

8 We do Choose the right exit route

Shared We do time. Unabsorbed material continues through the gut towards faeces; kidneys produce urine by filtering blood. Some gut residues are changed by bacteria, so the model does not claim that all food remains unchanged.

Reveal: Unabsorbed material continues through the gut. Kidneys make urine by filtering blood.

9 Hinge A water molecule is in the small intestine

A is incorrect: water does not need digestion into smaller food molecules. B is correct: water can be absorbed across the gut wall. C is incorrect: there is no direct intestine-to-bladder tube. Revisit the absorption diagram.

A. It must be digested before absorption. - Water does not need digestion into smaller food molecules.

B. It can be absorbed across the gut wall. - Water is absorbed in the small and large intestine.

C. It goes through a direct pipe to the bladder. - Kidneys make urine from blood; there is no direct stomach-to-bladder pipe.

10 You do Create the delivery explanation

10 minutes across You do. Support route provides words; standard explains order; stretch repairs ambiguity and includes model limitation.

11 You do Use three different verbs

Shared time. A pupil may point to a verb card and dictate an explanation. These verbs distinguish processes rather than competing systems.

12 You do Check the substance at each arrow

Shared time. Never accept “food goes into blood” without clarifying a small absorbed substance. Water need not be digested.

13 Review Trace and challenge

4 minutes across Review. Offer private check. A diagram with appropriate verbs can demonstrate equivalent understanding to prose.

14 Review Choose a closer look

Lundy: teacher records pupil questions, selects one magnified view and explains how pupil reasoning determined the next illustration.

15 Exit Three steps, three meanings

E1: absorption means crossing a surface into the body’s transport systems. E2: blood or plasma. E3: kidneys. E4 on the pupil page: water does not need digestion before absorption.

Answers and assessment

R1-R3

R1 breakdown of some food substances into smaller absorbable substances. R2 blood. R3 lungs.

W1-W3

W1 large starch needing digestion: digestion; glucose at gut wall: absorption; glucose in blood: transport; water in the gut: absorption. W2 water is absorbed from intestine into body transport; blood reaches kidneys which make urine carried to bladder. Not all swallowed water becomes urine. W3 unabsorbed material continues through large intestine to faeces; kidneys make urine from blood.

Hinge A-C

A incorrect: water does not need digestion. B correct: it can cross the gut wall. C incorrect: kidneys filter blood rather than a direct pipe from stomach.

S1-S6

S1 digestion. S2 absorption; blood. S3 no. S4 intestine→blood→tissues, with absorption at gut wall. S5 kidneys; gut. S6 digestion breaks some food substances down; transport carries absorbed substances onwards.

N1-N4

N1 digestion/breakdown; absorption/crossing surface into body transport; transport/carrying onwards. Glucose or water crossing gut wall is a valid example. N2 glucose crosses small intestine wall into blood and is carried towards tissues. Full correct liver/heart/lung route may be added but not required. N3 water absorbed into blood and distributed; already a small substance, not needing digestion. N4 gut residues→large intestine→rectum/anus/faeces; blood→kidneys→ureters/bladder→urine route. Names beyond kidneys/bladder optional.

T1-T4

T1 some food molecules are digested into small substances such as glucose, absorbed through the gut wall and transported in blood. T2 absorption precedes distribution; kidneys regulate water by filtering blood; water also remains in the body or leaves by other routes. T3 most dietary fats initially enter lymph rather than directly blood; label "a soluble nutrient such as glucose" instead of every nutrient. T4 gut wall is living tissue with cells and transport processes, not mesh holes; arrows represent substance movement, not whole food crumbs.

E1-E4 / reflection

E1 movement across a surface into the body's transport systems. E2 blood/plasma. E3 kidneys. E4 no; water can be absorbed without breakdown into smaller food units. Reflection: any reasoned magnification question.

Assessment and re-teach

Secure: distinguishes three processes, describes glucose/water absorption and blood transport, and separates kidney urine from gut residue. If vocabulary alone is correct, require a substance arrow. If direct-bladder error persists, trace the two parallel waste routes with different tokens.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

This lesson zooms in on crossing surfaces and transporting substances; the previous heart lesson supplies the detailed circulation route. Diagrams omit the liver, heart and lungs in some job maps and explicitly state that limitation.

Most dietary fat takes a lymphatic route before joining circulation. Core pupils follow glucose; stretch evaluates an “all nutrients directly to blood” claim.

Interactive route predictions map to W1-W3. On paper, move labelled tokens and explain each process aloud or in writing; no literal sieve experiment is proposed.

Access and responsive teaching

Supported

Use route words and sort tokens with sentence frames.

Standard

Explain two substance journeys and distinguish waste routes.

Stretch

Repair a delivery model and explain simplification and uncertainty.

Regulation

Offer quiet pointing, private writing, drawing or adult scribing. Pass-and-return for public questions. Use fictional cases; no personal medical disclosure or comparison of bodies. Assign response routes from current evidence, not fixed pupil labels.

Misconceptions to check

Absorption and digestion are the same job.

Digestion breaks some food substances down; absorption moves substances across the gut wall.

Check: Ask which job changes food molecules and which moves substances across a surface.

Water goes from stomach directly to bladder.

Water is absorbed into the body; kidneys filter blood and make urine.

Check: Ask where the kidneys get the water they remove.

All food enters the bloodstream unchanged.

Small absorbable substances cross the gut wall; unabsorbed material continues through the digestive tract.

Check: Ask where a whole bread crumb belongs.

Word	Meaning
absorption	The movement of substances across a surface into the body’s transport systems.
soluble	Able to dissolve in a liquid.
glucose	A small sugar the body can use in respiration.
plasma	The liquid part of blood, mostly water.
kidney	An organ that filters blood and helps balance water and dissolved substances.

Scientific references

NIDDK: Your digestive system and how it works

Checked September 2026. Digestion, absorption, transport and water absorption. Paraphrased factual reference; diagrams and activities are original.

NIDDK: Your kidneys and how they work

Checked September 2026. Kidneys filter blood and regulate water; urine does not come directly from a stomach-to-bladder tube.

NHLBI: How blood flows through the heart

Checked September 2026. Right-heart/lung/left-heart/body sequence and oxygenation. No adult pulse thresholds are used for pupils.